[HRD FINAL PROJECT]

HRD Intelligence
System Overview and Scope (Draft v5)

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Contents

1. User Roles..... 3
    1.1 HRD Admin..... 3
    1.2 Instructor..... 3
2. Core Features..... 4
    2.1. Authentication (Keycloak: Admin, Instructor)..... 4
    2.2. Student Management & Attendance..... 4
    2.3. Alumni Management..... 5
    2.4. Data Management (Admin & Instructor)..... 6
    2.5. Dashboard & Analytics..... 6
    2.6. Notifications..... 7
    2.7. AI Chatbot..... 7
    2.8. Autonomous AI Agents (Background Automation)..... 8
3. Matrix of Web and App Features..... 8
4. System Architecture..... 9
5. AI & Intelligent Processing..... 10
    5.1 Data Integrations..... 10
    5.2 AI Components:..... 10

HRD Intelligence – AI-Powered HRD Management Platform

HRD Intelligence is an AI-powered management platform designed to transform HRD data into actionable insights. The platform combines centralized HRD data management, automation, analytics, and a **Local LLM** to help HRD administrators and instructors manage students, alumni, attendance, academic performance, allowances, feedback, and operational reports more efficiently.

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1. User Roles

The HRD Intelligence system supports three main user roles: **HRD Admin**, **Instructor**, and **Alumni**. Each role has different permissions and access to information based on their responsibilities.

1.1 HRD Admin

- View student information, attendance, scores, academic records, and extra-class requests
- Manage extra-class hours, allowance calculations, and payment records
- Monitor student attendance and academic performance
- Manage HRD data and documents, including importing, reviewing, validating, and organizing records
- Generate operational and management reports, including attendance, scores, extra-class hours, allowances, and alumni information
- Use AI to analyze HRD data and generate insights
- Use the AI chatbot to ask questions about HRD administration and authorized HRD information, including:
  - Student information
  - Attendance and academic performance
  - Extra-class activities and allowances
  - Alumni profiles and employment information
  - HRD policies and general information
- Manage academic warnings and notifications
- Receive warnings when students may require academic or administrative attention

1.2 Instructor

- Monitor student academic progress
- Submit private feedback and opinions about students, including behavior, soft skills, hard skills, and academic performance
- View AI-generated summaries of instructor feedback
- Receive overtime and extra-class reminders
- Submit and view overtime reports
- Use the AI chatbot to ask questions and receive summaries

2. Core Features

2.1. Authentication (Keycloak: Admin, Instructor)

- **Login**
  - Email
  - Password
- **Forgot Password**
  - Enter email
  - Receive a password reset link.
- **Change password**
  - Current password
  - New password
  - Confirmation.

2.2. Student Management & Attendance

- **Admin + Instructor:**
  - **Monitor Student Profiles:**
    - View academic performance and personal profiles.
    - View students information: Name, Contact (email and phone), Gender, Date of Birth, Generation, Classroom, Course, Enrollment Date, and University.
    - Search students by name and email.
    - Filter students by generation, class, course, and status (active, dropped, graduated, all).
    - Can view instructor feedback about student
  - **Monitor Attendance**:
    - View student attendance records.
    - Filter attendance by class, date (month or specific day), and status (Present, Absent, Late, Permission).
    - Search attendance by student name.
  - **Monitor Academic Scores:**
    - View student scores (assignments, quizzes, exams, and homework)
    - The system automatically calculates total and average scores.
    - Search scores by student name
    - Filter by classroom or score range.
  - **At-Risk Student Tracking:**
    - Automatically identify students who may be at risk based on low attendance and/or declining academic performance.
    - Generate **alerts** based on defined risk indicators and performance trends.
    - Provide **AI-suggested support** actions based on the available student data.
- **Admin-Only Capabilities:**
  - **Extra-Class & Allowance Management:**
    - View students' extra-class records and total extra-class hours.
    - Automatically calculate extra-class allowance based on recorded hours.
    - Manage and update extra-class allowance per hour.
    - View allowance history by student, class, or monthly period.
  - **Monthly Allowance Management:**
    - Manage and update monthly allowance types and amounts, including:
      - Meal allowance
      - Korean language allowance

        - IT score-based allowance
        - Extra-class allowance
        - Class leader allowance
        - Attitude and attendance allowance
        - Other HRD-provided allowances
    - Manage IT score-based allowance ranges (e.g., ≥90, ≥80).
    - Update allowance rules and amounts when policies change.
    - Automatically calculate each student's total monthly allowance.
    - Export monthly allowance reports (CSV, Excel)

- **Instructor:**
    - **Instructor Feedback**
        - Submit private feedback about students (student name, class, course, feedback, date)
        - **Categorize feedback** by Behavior, Soft Skills, Hard Skills, Other observations
        - View, edit, and manage previously submitted feedback.
        - Filter feedback by class, course
        - Student feedback is private and is not visible to students

2.3. Alumni Management

- **Admin** :
    - **Profile Management**: Create, view, and edit alumni personal info, education, generation, course, salary range, phone number, and uploaded documents.
    - **Search & Filter:**
        - Search by name or email.
        - Filter by generation, course, employment status (employed, self-employed, studying, unemployed), industry, salary range, and education.
    - **Profile Update Requests:**
        - **Manually or automatically (via AI) send an update request form directly to an alumnus's email. *(When the alum fills out this emailed form, it updates their data in the Admin system)*.**
    - **Employment Tracking**:
        - Track current employment status (employed, self-employed, studying, unemployed)
        - Maintain a history of past and current jobs (company name, job title, industry, and work location).
    - **Analytics:** View Alumni Statistics within the management module.

2.4. Data Management (Admin & Instructor)

- **Supported File Formats**
  - Users can upload and manage: TXT, MD, Excel (XLSX), CSV, PDF, Word (DOCX), PowerPoint (PPTX), and Images (PNG, JPEG).
- **Folder and File Management**
  - Create, Rename, Move, Delete and Duplicate folder/file
  - Export folder/file as ZIP
- **Ownership & Access Control**
  - Users can own documents and folders they create or upload.
  - The user who creates a folder becomes its **Owner**.
  - Owners can:
    - Share documents and folders
    - Manage access and permissions
    - Change user permissions
    - Permanently delete items
  - Supported permissions:
    - **Viewer** — Read-only access
    - **Editor** — Read and edit access
  - Users can access only:
    - Documents and folders they own
    - Documents and folders shared with them
- **Search & Organization**
  - Search files and folders by name.
  - Filter files by extension and owner
  - Sorted files and folders alphabetically, upload date

2.5. Dashboard & Analytics

- **Admin + Instructor:**
  - **Student Overview**
    - Total students
    - Active, Dropped, and Graduated students
    - Key metrics using KPI cards
  - **Student Analytics**
    - Gender distribution
    - Top 5 students by: Course score, Extra-class hours, IT score, Korean score
    - Both items can be filtered by Generation
  - **Student At Risk** (Students requiring attention)
    - Academic performance
    - Attendance rate
    - Status: Low, Medium, High
  - **Attendance Analytics**
    - Present, Absent, Late, Permission (Donut Chart)
    - Attendance rate over time (Line Chart) TStby Daily, Weekend, Monthly
- **Admin Only**
  - **Generation Overview** : Students by generation (Bar Chart) -> improve performance
  - **Alumni Overview** : Total alumni (KPI Card)
  - **Employment Status** : Employed, Unemployed, and Other (Horizontal Bar Chart)
  - **Reports** : Export dashboard reports that filter by week, month, or custom date range as CSV or PDF. -> can customize, chose fill about student and the preview before export

2.6. Notifications

- **Notification Channels:**
  - **In-App:**
    - Shows notifications inside the web and mobile applications.
    - Show unread notification count
  - **Email:**
    - Sends reminders, requests and status updates to the user's registered email.
- **Trigger Events by Role:**
  - **Admin + Instructor:**
    - Low Attendance: When a student's attendance falls below the defined threshold.
    - Declining academic: when a student's academic performance shows a declining trend.
  - **Admin Only:**
    - Alumni Updates: Notified in-app or via email when an alumnus completes and submits the emailed profile update form.
  - **Instructor Only:**
    - Overtime reminders: Notify instructors of extra-class requests on their standby days.

2.7. AI Chatbot

- **Core AI Capabilities:**
  - **Language** : English
  - **Input Type** : Text and Voice (Speech-to-Text).
  - **Technology** : Retrieval-Augmented Generation (RAG) + Local LLM.
  - **Actions** : Answer queries, Generate summaries/insights, Write/send emails (requires user confirmation)
  - **Output Formats : -> check**
    - Text, Files (TXT, MD, PPTX, DOCX, PDF),
    - Charts (Bar, Line, Donut, Data Tables exported to image formats).
  - **Chat History:**
    - Stores recent 10 sessions.
    - Users can search, rename, delete, or continue past conversations.
- **General AI Chat (Admin & Instructor):**
  - Acts as a general HRD assistant utilizing all authorized user data across multiple folders.
- **Folder-Specific AI Chat:**
  - Create isolated chat sessions within specific folders.
  - AI answers strictly from the files selected within that folder.
  - Can generate slides from folder documents (e.g., lesson notes).

2.8. Autonomous AI Agents (Background Automation)

- **Scheduled Agents (Run on a timer)**
  - **Weekly Summary** : Sends Admins a Friday email covering student attendance, grades, and extra hours.
  - **Feedback Summary**: Creates a weekly report from instructor notes about student progress and issues.
  - **Alumni Profile** : Emails alumni every 6 months to ask for their latest job details.

- **Trigger-Based Agents (Run when a specific event happens)**
  - **At-Risk Agent** : Drafts a warning email for the Admin if a student's attendance or grades drop.
  - **Alumni Job Alert** : Notifies Admins immediately when an alumnus submits the external form with updated career or industry info.

3. Matrix of Web and App Features

Capability	Admin Web	Instructor Web	Admin Mobile	Instructor Mobile
Authentication	Full	Full	Full	Full
Student management	Full	View	View	View
Attendance	Full	View	View	View
Scores	Full	View	Quick / View	View
Manage Extra Class	Full	—	Quick / View	—
Allowance	Full	—	Quick / View	—
Feedback	Full	Full	View	View / Submit
Alumni	Full	—	View	—
Data Management	Full	Upload / View / Manage Teaching Materials	Full View	View Teaching Materials
Notification	Full	Students Risk Alert / OT Reminder	Full Alert	Full Alerts
AI Chatbot	Full	Authorized	Full	Authorized

4. System Architecture

Layer	Component	Technology / Tool	Purpose
Web Frontend Next.js + TypeScript	UI & Styling	Tailwind CSS, shadcn/ui, Heroicons	UI and styling
	Forms	React Hook Form + Zod	Form handling and validation
	API	Axios + REST API	Backend communication
	Authentication	Keycloak, JWT	Authentication and authorization
	Charts & Visualization	Recharts	Charts, KPI cards, tables, and heatmaps
	Notifications	In-App + Email	System notifications
Android Kotlin + Jetpack Compose	Architecture	MVVM + StateFlow + Hilt	Architecture and dependency injection
	Networking	Retrofit + OkHttp	API communication
	Security	Keycloak + JWT + Android Keystore	Authentication and secure storage
	Notifications	Firebase Cloud Messaging	Push notifications
iOS Swift + SwiftUI	Architecture	MVVM + Observable	Architecture and state management
	Networking	URLSession + Codable	API communication
	Security	Keycloak + JWT + Keychain	Authentication and secure storage
	Database	SwiftData	Local data storage
Backend Java + Spring Boot	Architecture	Spring Cloud Microservices	Microservice architecture
	API Gateway	Spring Cloud Gateway	API routing
	Service Discovery	Eureka	Service discovery
	Configuration	Spring Cloud Config	Centralized configuration
	Security	Keycloak + Spring Security	Authentication and RBAC
	Database	PostgreSQL + Spring Data JPA	Data storage and access
AI Service Python + FastAPI	RAG	Chunking + Embeddings + Vector Search + LLM	Document-based AI responses
	LLM Inference	vLLM	Local LLM inference
	Embedding	BAAI/bge-m3	Text embeddings
	Vector Database	pgvector	Vector storage and search
	OCR	PaddleOCR / Tesseract	Document text extraction
	Document Processing	PyMuPDF + pandas + openpyxl	PDF, Excel, and CSV processing
	AI Analytics	pandas + Statistical Analysis	Data and trend analysis
	AI Chatbot	Semantic Search + RAG + LLM	HRD data interaction
Infrastructure & DevOps	Containerization	Docker	Application containerization
	CI	Jenkins	Build, test, and Docker image creation
	CD / GitOps	Argo CD + GitHub	Continuous deployment
	Orchestration	Kubernetes	Container orchestration and management
Analytics	Visualization	Recharts + Python/pandas	Data visualization and analysis
	Analytics Methods	Descriptive, Trend, Comparative, Correlation, Risk, Pattern, AI-assisted	Data analysis

5. AI & Intelligent Processing

The platform uses a **Local LLM architecture** to process HRD data while keeping organizational data within the local/private environment.

5.1 Data Integrations

- **RDA API** : Integrates student attendance, student name, check-in, and check-out data.
- **Assessment API** : Integrates student information, scores, staff details, class, course, organizational chart, and related metrics.

5.2 AI Components:

- **Local LLM**
  - Natural-language Q&A
  - Data analysis
  - Summarization
  - Insight generation
- **RAG**
  - Retrieve relevant HRD information
  - Ground LLM responses in organizational data
  - Support document-based Q&A
- **Embeddings + Vector Database**
  - Convert documents and data into semantic representations.
  - Store embeddings for retrieval.
  - Enable similarity search for RAG.
- **OCR & Document Processing**
  - PaddleOCR
    - Extract text from scanned documents and images.
    - Support text detection and recognition.
    - Provide confidence scores for OCR results.
    - Support multilingual documents.
    - Process forms, reports, academic records, attendance documents, and other HRD documents.
    - Run locally without sending sensitive documents to external cloud services.
  - PP-StructureV3
    - Detect document structure such as titles, paragraphs, tables, and content regions.
    - Preserve document structure for downstream processing.
    - Improve document chunking by keeping related content grouped together.
    - Convert extracted content into structured formats such as Markdown or JSON.
    - Support complex document processing for RAG.
  - Tesseract
    - Alternative OCR engine for text extraction.
    - Used when appropriate for supported document types